THE UPPER SHRIEK IS COMPUTED BY THE DUALITY INVOLUTION APPLIED TO THE PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AU2 (sources/stacks-duality/pinned-original-source.tex line 4734).

Allowed standard prerequisites (editorial).

- The right adjoint a of pushforward along a compactification and the description f^! = j^* o a. Same chapter, Lemma labelled lemma-twisted-inverse-image, cited at proof lines 4750-4754. Reduces the statement to the compactified case.
- The duality involution on the bounded coherent derived category of a Noetherian scheme with a dualizing complex. Same chapter, Lemma labelled lemma-dualizing-schemes, cited at proof lines 4785-4787. Gives D_Y(D_Y(M)) = M, the key step of the chain.
- The internal Hom adjunction and its behaviour under the tensor product with a pullback. Cohomology, Lemma labelled cohomology-lemma-internal-hom, cited at proof lines 4764-4766, 4776-4778 and 4791-4793. Turns the Hom groups into tensor products and back.
- Projection formula style base change for the cohomology of a tensor product. Derived Categories of Schemes, Lemma labelled perfect-lemma-cohomology-base-change, cited at proof lines 4770-4772. Moves the pullback out of the pushforward.
- Nagata compactification. More on Flatness, Theorem labelled flat-theorem-nagata and Lemma labelled flat-lemma-compactifyable, cited at proof lines 4748-4750. Provides the compactification used throughout.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original theorem 0F41: sources/stacks-chow/tags.txt, source lines 18886–18886. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4750 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9Z: sources/stacks-chow/tags.txt, source lines 12967–12967. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4751 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line 4752 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08DJ: sources/stacks-chow/tags.txt, source lines 10639–10639. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4785 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08EU: sources/stacks-chow/tags.txt, source lines 10685–10685. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4788 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A89: sources/stacks-duality/pinned-original-source.tex, source lines 472–472. Same-chapter pointer into the author environment lemma at source line 472; invoked at proof line 4791 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. Let K be a dualizing complex on Y. Set $D_Y(M) = R\mathcal{H}om_{\mathcal{O}_Y}(M, K)$ for $M \in D_{Coh}(\mathcal{O}_Y)$ and $D_X(E) = R\mathcal{H}om_{\mathcal{O}_X}(E, f^!K)$ for $E \in D_{Coh}(\mathcal{O}_X)$. Then there is a canonical isomorphism*

$$f^!M \longrightarrow D_X(Lf^*D_Y(M))$$

for $M \in D^+_{Coh}(\mathcal{O}_Y)$.

Proof. Choose compactification $j : X \subset \overline{X}$ of X over Y (More on Flatness, Theorem 0F41 and Lemma 0A9Z). Let a be the right adjoint of Lemma 0A9E for $\overline{X} \to Y$. Set $D_{\overline{X}}(E) = R\mathcal{H}om_{\mathcal{O}_{\overline{X}}}(E, a(K))$ for $E \in D_{Coh}(\mathcal{O}_{\overline{X}})$. Since formation of $R\mathcal{H}om$ commutes with restriction to opens and since $f^! = j^* \circ a$ we see that it suffices to prove that there is a canonical isomorphism

$$a(M) \longrightarrow D_{\overline{X}}(L\overline{f}^* D_Y(M))$$

for $M \in D_{Coh}(\mathcal{O}_Y)$. For $F \in D_{QCoh}(\mathcal{O}_X)$ we have

$$\begin{aligned}
\mathrm{Hom}_{\overline{X}}(F, D_{\overline{X}}(L\overline{f}^* D_Y(M))) &= \mathrm{Hom}_{\overline{X}}(F \otimes^{\mathbf{L}}_{\mathcal{O}_X} L\overline{f}^* D_Y(M), a(K)) \\
&= \mathrm{Hom}_Y(R\overline{f}_*(F \otimes^{\mathbf{L}}_{\mathcal{O}_X} L\overline{f}^* D_Y(M)), K) \\
&= \mathrm{Hom}_Y(R\overline{f}_*(F) \otimes^{\mathbf{L}}_{\mathcal{O}_Y} D_Y(M), K) \\
&= \mathrm{Hom}_Y(R\overline{f}_*(F), D_Y(D_Y(M))) \\
&= \mathrm{Hom}_Y(R\overline{f}_*(F), M) \\
&= \mathrm{Hom}_{\overline{X}}(F, a(M))
\end{aligned}$$

The first equality by Cohomology, Lemma 08DJ. The second by definition of a. The third by Derived Categories of Schemes, Lemma 08EU. The fourth equality by Cohomology, Lemma 08DJ and the definition of D_Y. The fifth equality by Lemma 0A89. The final equality by definition of a. Hence we see that $a(M) = D_{\overline{X}}(L\overline{f}^* D_Y(M))$ by Yoneda's lemma. □